The Invariance Principle — problem 1.23

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Start with n pairwise different integers $x_1, x_2, \ldots, x_n$, $(n > 2)$ and repeat the following step :

$$T : (x_1, \ldots, x_n) \mapsto \left(\frac{x_1 + x_2}{2}, \frac{x_2 + x_3}{2}, \ldots, \frac{x_n + x_1}{2} \right).$$

Show that $T, T^2, \ldots$ finally leads to nonintegral components.